

Poly (4-styrenesulfonic acid): A recoverable and reusable catalyst for acid hydrolysis of polyethylene terephthalate

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ABSTRACT

Poly (4-styrenesulfonic acid) (PSSA) was used as a recoverable acid catalyst for the hydrolysis of polyethylene terephthalate (PET) to terephthalic acid (TPA) and ethylene glycol (EG). The effect of catalyst concentration, time, and temperature on the depolymerization of PET and yield of TPA were determined with PSSA and H₂SO₄. The PSSA system exhibited faster hydrolysis kinetics than H₂SO₄ with a shorter induction period than those often observed for acid hydrolysis of PET. The activation energies for the hydrolysis of PET were 24.8 kJ mol⁻¹ and 29.1 kJ mol⁻¹ for 2 M PSSA and 2 M H₂SO₄, respectively. The hydrophobic backbone of PSSA improves surface wetting of the catalyst solution at the PET interface, resulting in a higher localized concentration of protons. Contact angle measurements indicated better surface wetting of PET films by the aqueous solutions of PSSA than H₂SO₄ at equal concentrations. Additionally, the reutilization and titration results indicated that PSSA maintained the activity after reuse for five reaction cycles.

1. Introduction

Polyethylene terephthalate (PET) is a semi-crystalline, colorless polyester with excellent thermal and mechanical properties that is readily processed into films, fibers, and bottles [1]. Because of these unique properties, PET has found broad applications, including food and beverage packaging and in textiles. The annual growth rate for the global PET market is estimated to be around 4.2% over the five-year period from 2019 to 2024, increasing from 40 billion US dollars in 2019 to 51 billion US dollars in 2024 [2]. While PET is the most recycled polymer by volume, the recycle rate for PET in the US is less than 30% [3]. The broad use of PET combined with a low recycle rate results in accumulation of high volumes of waste PET in landfills and the environment [4–9].

While most of the commercial recycling of PET is through mechanical recycling, this method requires clean polymer streams and can lead to degradation in properties, thus, limiting the volume of recycled PET incorporated into the process streams [10,11]. Chemical recycling can be used to depolymerize the polymer and produce monomers or upgraded molecules for broad industrial applications [12–14]. A variety of processes have been explored for chemical recycling of PET including

glycolysis, methanolysis, and hydrolysis [15–24]. Hydrolysis produces terephthalic acid (TPA) and ethylene glycol (EG) that can be directly polymerized to virgin PET. Alkaline hydrolysis has been reported in presence of base catalysts, such as sodium hydroxide and potassium hydroxide in aqueous solutions [25,26] and non-aqueous systems [27–29]. In contrast, neutral hydrolysis typically takes place in water or steam under pressure and at high temperatures, in the range of 200 °C–450 °C [30–32].

Acid hydrolysis is a well-established process in which the reactions occur at moderate temperatures but require high acid concentrations and long reaction times [33–37]. The important factors that determine the extent of PET conversion and TPA yield are PET particle size, catalyst type and concentration, reaction temperature, and reaction time. For example, Yoshioka et al. [38] converted more than 90% of PET using a 7 M sulfuric acid (H₂SO₄) aqueous solution at 150 °C for 5 h. A modified shrinking-core model was developed to describe the kinetics of the reaction [39]. Nitric acid [40] and hydrochloric acid [41,42] have also been reported as other acid catalysts for PET recycling. The main drawback of acid hydrolysis is the use of high molarity acid solutions that require costly equipment to avoid corrosion and a subsequent neutralization step after reaction.

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Since hydrolysis occurs at the PET surface, wetting by the catalyst solution and adsorption of the catalyst may be important in determining the reaction kinetics [38–40]. For example, Amarasekara et al. demonstrated that the reaction kinetics for acid hydrolysis at the surface of cellulose increased for alkyl sulfonic acid. The reaction rate constant correlated well with the distribution coefficients of these alkyl sulfonic acid catalyst between octanol and water [43]. Poly (4-styrenesulfonic acid) (PSSA) is a soluble catalyst that combines the rapid kinetics of H_2SO_4 with ease of recovery and reuse. PSSA has been used in different reactions for biomass conversion, including biodiesel synthesis, xylose dehydration to furfural, hydroxymethylfurfural production from fructose, and the selective oxidation of furfural with H_2O_2 to C_4 diacids [44–47]. In addition, PSSA in aqueous medium has been shown to adsorb on hydrophobic surfaces (e.g., alumina) [48–51].

Here, we report the use of PSSA as a recoverable acid catalyst for the hydrolysis of PET along with a comparison with H_2SO_4 . Specifically, we have studied the impact of the reaction temperature, catalyst concentration, and reaction time on PET conversion and TPA yield. PSSA exhibited increased hydrolysis kinetics relative to H_2SO_4 over the full range of conditions investigated. This improvement was attributed to increased surface wetting by the PSSA solutions combined with the adsorption of PSSA on the PET surface, which would increase the local concentration of Brønsted acid sites (H^+) for hydrolysis.

2. Experimental

2.1. Materials

Poly (sodium 4-styrenesulfonate) (PSSS) with an average molecular weight of 70 kDa, terephthalic acid (TPA), ammonium hydroxide solution (30–33% NH_3 in H_2O), methyl sulfoxide- d_6 , and deuterium oxide were all purchased from Sigma-Aldrich. Sulfuric acid (Certified ACS plus) was purchased from Fisher Scientific. Amberlyst®-15 was purchased from Alfa Aesar and was ground to a fine powder for the ion-exchange of PSSS to PSSA. PET pellets were generously donated by Alpek and were grounded and sieved to a powder in the diameter size range of 250–354 μm .

2.2. PSSA catalyst synthesis

PSSS was converted into the protonated form by ion exchange using a sulfonic resin, as reported elsewhere [44]. Briefly, 100 mL of PSSA (25 wt%) solution was added to the mixture of 120 g of grounded Amberlyst®-15 and 500 mL of DI water. Then, it was stirred at room temperature overnight. After ion exchange, the mixture was filtered to separate the resin from the solution, and the PSSA solution was dried at 60 °C to recover the PSSA. The recovered PSSA films were cut into small pieces and dried for another 6 h at 60 °C to minimize the water content. Thermogravimetric analysis (TGA) of the recovered PSSA indicated that the synthesized PSSA had 4.2% water (Fig. S1 in the Supplementary Material). A complete ion exchange would generate a PSSA containing 5.4 mmol H^+ /g. As the titration results showed an acidity of 4.7 mmol H^+ /g, this indicated 87% completion of the ion exchange.

2.3. Titration measurements

Acid-base titration was performed to determine the proton concentration on PSSA. In a typical titration test, around 20 mg of PSSA was dissolved in 10 mL DI water. The solution was titrated against a 0.01 N NaOH solution standardized with potassium acid phthalate. To determine the end point, phenolphthalein was used as the indicator. Each titration was repeated three times.

2.4. PET hydrolysis reactions

For the PET hydrolysis reactions, 0.2 g of PET powder was placed in a

15-mL Ace pressure reactor equipped with a magnetic stirrer and 10 mL of 1, 2 or 4 M PSSA aqueous solutions. The solutions were prepared by dissolving PSSA in DI water at temperatures up to 50 °C. The reactor was immersed in a preheated oil bath at 130, 140, and 150 °C. The same reaction conditions were used for 1, 2, and 4 M H_2SO_4 solutions in order to maintain constant concentrations of catalyst. Every reaction time was an independent measurement and was repeated three times.

After a predefined time, the reaction was quenched by submerging the reactor in a cold-water bath, and the unreacted PET and TPA powders were recovered by filtration. The TPA product was separated from unreacted PET by dissolving the mixture in a 5 M NH_4OH solution [33]. The recovered TPA was compared with commercial TPA by Fourier-transform infrared spectroscopy (FTIR) and proton nuclear magnetic resonance (^1H NMR) analyses (Fig. S2 and Fig. S3 in the Supplementary Material). After that, PSSA was recovered for further uses by evaporating the filtrate solution. The recovered PSSA was washed several times with an excess of water to ensure the removal of EG and dried to remove water that would affect the mass of catalyst.

The PET conversion (%) was determined by Equation (1):

$$\text{PET Conversion (\%)} = \left(\frac{m_{\text{PET},i} - m_{\text{PET},f}}{m_{\text{PET},i}} \right) \cdot 100 \quad (1)$$

where $m_{\text{PET},i}$ is the initial mass of PET (0.2 g), and $m_{\text{PET},f}$ is the final mass of dried PET.

The molar yield of TPA (TPA yield (%)) was determined by Equation (2):

$$\text{TPA Yield (\%)} = \left(\frac{m_{\text{TPA}}}{m_{\text{PET},i}} \right) \left(\frac{MW_{\text{PET}}}{MW_{\text{TPA}}} \right) \cdot 100 \quad (2)$$

where m_{TPA} , MW_{PET} , and MW_{TPA} represent the mass of recovered TPA, molecular weight of PET (192.2 g mol^{-1}), and molecular weight of TPA (166.1 g mol^{-1}), respectively.

2.5. Characterization of recovered PSSA and TPA

FTIR and ^1H NMR were used to compare the structure of the recovered TPA with that of commercial TPA. Also, the as-synthesized PSSA and the recovered PSSA were analyzed by FTIR, ^1H NMR, and ^{13}C NMR for structural comparison. For FTIR, the samples were mixed with KBr, pressed into thin pellets and analyzed in a Varian Excalibur Series FTIR spectrometer with microscope transmission mode in the range of 400–4000 cm^{-1} . ^1H NMR spectra were recorded in a Bruker Avance III spectrometer at 600 MHz. To do so, the TPA samples were dissolved in methyl sulfoxide- d_6 and the PSSA samples were dissolved in deuterium oxide.

2.6. Contact angle measurements

Contact angle measurements [52] were used to determine the relative wetting of the surface of PET films by PSSA and H_2SO_4 aqueous solutions with concentrations from 0.125 to 2 M at room temperature. PET films of 0.35 mm average thickness were processed in a HAAKE Rheomex Brabender single screw extruder for this study. The contact angle was measured using a Keyence VHX-600 digital microscope equipped with a camera in the reflection mode with a full-coaxial light stage. For a typical test, a droplet of PSSA or H_2SO_4 solution was placed on a PET film affixed to a flat glass substrate. An image of the droplet on the PET surface was captured to determine the contact angle. Each test was repeated at least six times by placing new droplets on a new film.

2.7. Ethyl acetate hydrolysis (model system)

Briefly, 5 mL of a 5 wt% ethyl acetate solution was transferred to a 15-mL Ace pressure reactor followed by the addition of 5 mL of a 0.5 M H_2SO_4 solution. Then, the reactor was immersed in a silicon oil bath

stirred at 600 rpm. Because of how fast the hydrolysis process occurs, the reaction was carried out for 1 h at temperatures ranging from 40 °C to 70 °C. Upon completion, the reactor was immersed in crushed ice to stop the reaction. To determine the extent of the reaction (production of acetic acid) and the activation energy, 2 mL of the reaction sample was titrated against a 0.1 N NaOH solution standardized with potassium hydrogen phthalate, and the H^+ content of the catalyst was subtracted from the total. For the sake of comparison, similar reactions were conducted using 5 mL of PSSA solutions with the same H^+ content as the H_2SO_4 solutions. To do so, it was taken into account that a 0.5 M H_2SO_4 solution dissociates mostly as H^+ and HSO_4^- (instead of $2H^+$ and SO_4^{2-}) [53]. Finally, the activation energies of the hydrolysis of ethyl acetate with PSSA and H_2SO_4 were determined and compared.

3. Results and discussion

3.1. Temperature effect on the reaction kinetics

As PSSA has been shown to be thermally stable at temperatures equal to or lower than 150 °C, this work focused on studying the hydrolysis kinetics of PET at temperatures between 130 °C and 150 °C [47,54–56]. Fig. 1 and Fig. 2 show the effect of the temperature on the reaction kinetics in terms of PET conversion and TPA yield obtained with 2 M solutions of PSSA and H_2SO_4 , respectively. Note that each point was an independent PET hydrolysis experiment and was run in triplicate, with the error bars indicating the standard deviation. The PET conversions and TPA yields follow similar trends for all systems studied. It should also be noted that the TPA yield was slightly lower than the PET conversion, particularly at lower conversion, likely due to the loss of TPA

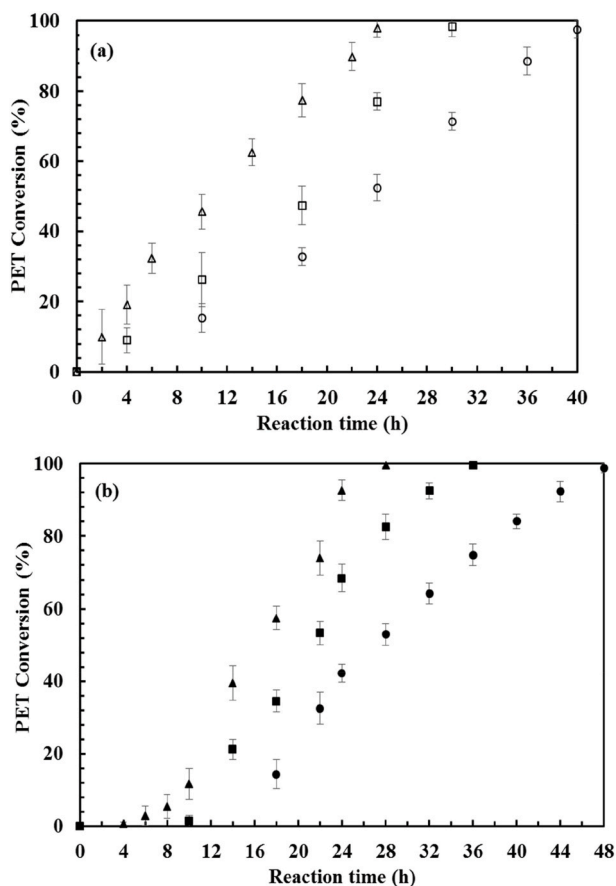


Fig. 1. Reaction temperature effect on the PET conversion (%) with 2 M solutions of (a) PSSA at 150 °C (Δ), 140 °C (□), and 130 °C (○), and (b) H_2SO_4 at 150 °C (▲), 140 °C (■), and 130 °C (●).

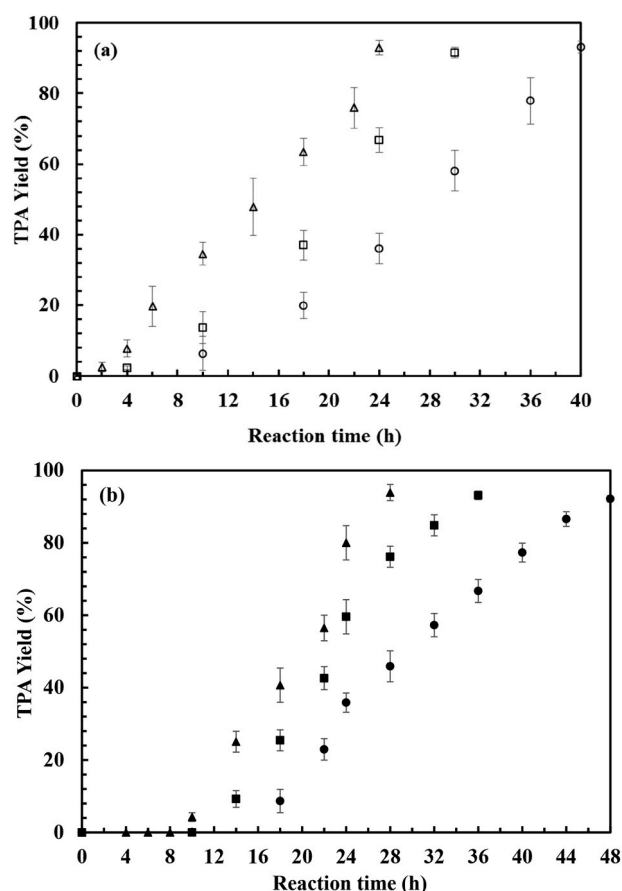


Fig. 2. Reaction temperature effect on the TPA yield (%) for 2 M solutions of (a) PSSA at 150 °C (Δ), 140 °C (□), and 130 °C (○), and (b) H_2SO_4 at 150 °C (▲), 140 °C (■), and 130 °C (●).

during the product recovery phase. Therefore, the TPA yield was used in fitting modeling to provide a conservative value for the rate constant for these systems.

As shown in Fig. 1b, a significant induction period was observed for this reaction when using H_2SO_4 as the catalyst for all temperatures studied. This induction time has been attributed in the literature to the formation of cracks in the PET surface by the acid catalyst that facilitates access to the interior of the PET particles [33,38,39]. The induction time was much shorter for the 2 M PSSA solution and decreased with increasing reaction temperature for both catalysts. As expected, there was a decrease in the reaction rate with decreasing reaction temperature for both catalysts.

The shrinking-core model, reported by Mancini et al. [33], was applied to determine the rate constants and activation energies with PSSA and H_2SO_4 catalysts. In this model, it was assumed that the TPA produced during the hydrolysis reaction did not deposit on the PET surface because of continuous stirring during the reaction [33,38,39]. The final equation for the TPA yield as a function of the reaction time (t), density of spherical PET particles (ρ_{PET}), catalyst concentration (C), and the rate constant per unit surface area (k) is:

$$\begin{aligned} (1 - [TPA Yield]_{induction})^{\left(\frac{1}{3}\right)} - (1 - [TPA Yield]_t)^{\left(\frac{1}{3}\right)} \\ = \left(\frac{k \cdot C}{\rho_{PET} \cdot r_i}\right)(t - t_{induction}) \end{aligned} \quad (3)$$

where $[TPA Yield]_{induction}$ and $t_{induction}$ are the amount of recovered TPA at the induction time and the required time to reach the linear part of the TPA yield vs t plot, respectively. $\left(\frac{k \cdot C}{\rho_{PET} \cdot r_i}\right)$ is the apparent reaction rate

constant (time^{-1} units), also denoted as K , with k as the specific reaction rate constant, C as the catalyst concentration (H^+), ρ_{PET} as the density of PET, and r_i indicating the initial size (radius) of the PET particles. The induction time was used to allow for better comparison between systems with broad differences in onset of PET degradation. Equation (3) was used together with TPA yield data in Fig. 2 to calculate the apparent reaction rate constant (K) with PSSA and H_2SO_4 catalysts at each reaction temperature. Then, the specific rate constants (k) were calculated from the apparent rate constants using the density of PET (1.38 g/cm^3) and an average radius of 0.015 cm for the PET particles. Table 1 presents the k values obtained from the hydrolysis reaction with 2 M solutions of PSSA and H_2SO_4 . For all cases, the k values obtained with PSSA were slightly higher than those with H_2SO_4 , indicating the higher activity of PSSA. The Arrhenius equation (Equation (4)) was used to determine the activation energies with PSSA and H_2SO_4 .

$$\ln(k) = \ln(k_0) - \frac{E_a}{RT} \quad (4)$$

where k is the specific reaction rate constant, k_0 is the pre-exponential factor, E_a is the activation energy, R is the universal gas constant, and T is the reaction temperature. Fig. 3 shows the fit of the model to the data from the hydrolysis reactions at 150°C with 2 M solutions of PSSA and H_2SO_4 . Fig. 4 shows the Arrhenius plots obtained with 2 M PSSA and 2 M H_2SO_4 as catalysts. The activation energies obtained with PSSA and H_2SO_4 were 24.8 kJ mol^{-1} and 29.1 kJ mol^{-1} , respectively. Activation energies for PET acid hydrolysis have been reported in the range between 57 kJ mol^{-1} with aqueous solutions of phosphoric and sulfuric acid to 101 kJ mol^{-1} for aqueous solutions of hydrochloric acid and nitric acid [34,40,57]. The lower activation energies here obtained may be due to the continuous stirring of the reaction systems during the hydrolysis of PET. The slightly lower activation energy obtained with PSSA is consistent with the increased activity relative to H_2SO_4 .

3.2. Catalyst concentration effect on acid hydrolysis reaction

Fig. 5 and Fig. 6 show the PET conversions and TPA yields obtained with different catalyst concentrations at 150°C . As seen in these figures, PET hydrolyzed more rapidly when using PSSA solutions than with H_2SO_4 for all catalyst concentrations studied. As expected, there was an increase in the reaction rate at higher catalyst concentration along with a decrease in the induction time for PET depolymerization. When using 1 M, 2 M, and 4 M solutions of PSSA, more than 90% PET conversion and TPA yield were achieved in 30, 24, and 16 h, respectively, while it took 36, and 28, and 18 h when using the same concentration of H_2SO_4 . The difference in time required to achieve a specific conversion was primarily due to the longer induction times when using H_2SO_4 particularly at lower concentrations. Table 2 shows the apparent rate constant obtained by using the shrinking-core model at the reaction temperature of 150°C . As expected, the apparent rate constants (K) were higher for higher catalyst concentrations, being these similar for the two catalysts studied.

3.3. Hydrolysis of ethyl acetate (model system)

Fig. 7 shows the Arrhenius plot obtained for the hydrolysis of ethyl acetate when using 0.5 M H_2SO_4 and 0.5 M PSSA as catalysts. As can be seen in Fig. 7, the activation energy obtained with H_2SO_4 (39.6 kJ

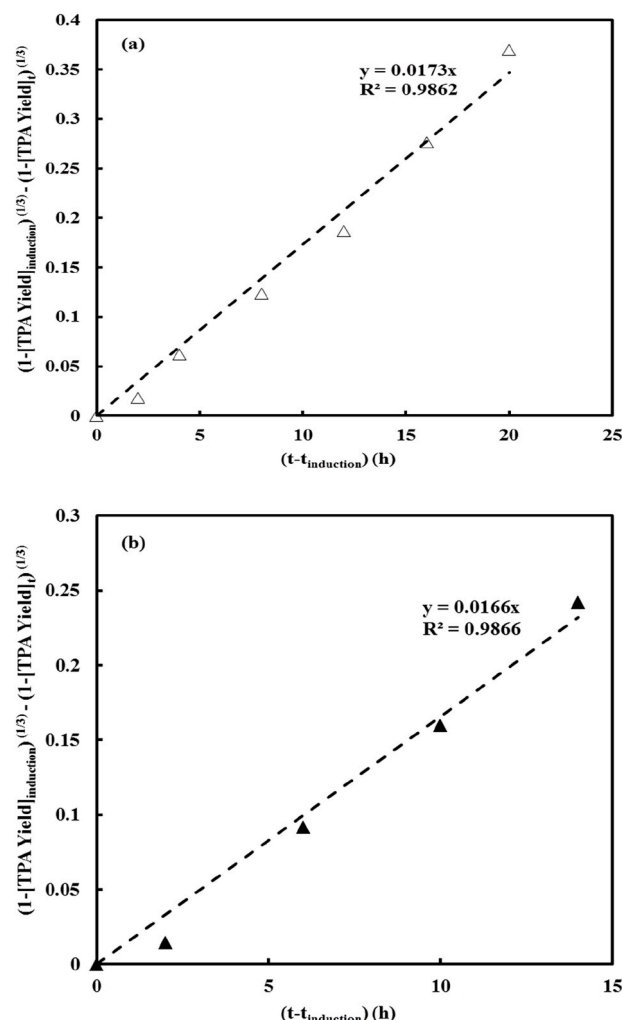


Fig. 3. Shrinking-core model fitting to the results obtained with 2 M solutions of (a) PSSA (Δ), and (b) H_2SO_4 ($\blacktriangle$) at $T_{\text{reaction}} = 150^\circ\text{C}$.

mol^{-1}) was considerably lower than that with PSSA (53.8 kJ mol^{-1}), which shows a faster hydrolysis rate in the presence of H_2SO_4 . These results are consistent with those in the literature, in which the activation energies for the hydrolysis of ethyl acetate range between 30 and 70 kJ mol^{-1} [58–60]. As H_2SO_4 is a mineral acid, all protons are readily available for the hydrolysis of ethyl acetate. In contrast, due to the nature of PSSA, there might be some active sites not exposed to the reactants, thereby reducing the catalytic activity. A similar performance was reported in the literature for the acid-catalyzed methanolysis of tributyrin, in which the authors used H_2SO_4 , PSSA, and solid sulfonic resins, and concluded that the activity of PSSA was lower than that of a pure homogeneous catalyst (H_2SO_4) but higher than that of heterogeneous catalysts (sulfonic resins), being this attributed to the accessibility of the reactants to the Brønsted acid sites [44]. Even when H_2SO_4 showed to be more active than PSSA in the hydrolysis of ethyl acetate, the opposite was observed in the case of the hydrolysis of PET (Fig. 4 and 7), which might indicate the occurrence of an additional important

Table 1
Specific reaction rate constants (k) obtained with 2 M aqueous solutions of PSSA and H_2SO_4 as a function of the reaction temperature.

	2 M PSSA	2 M H_2SO_4
T ($^\circ\text{C}$)	k ($\text{kg}_{\text{PET}}\cdot\text{m}\cdot(\text{mol catalyst})^{-1}\cdot\text{s}^{-1})\cdot(10^{10})$	k ($\text{kg}_{\text{PET}}\cdot\text{m}\cdot(\text{mol catalyst})^{-1}\cdot\text{s}^{-1})\cdot(10^{10})$
150	5.07	4.80
140	4.14	3.76
130	3.57	3.18

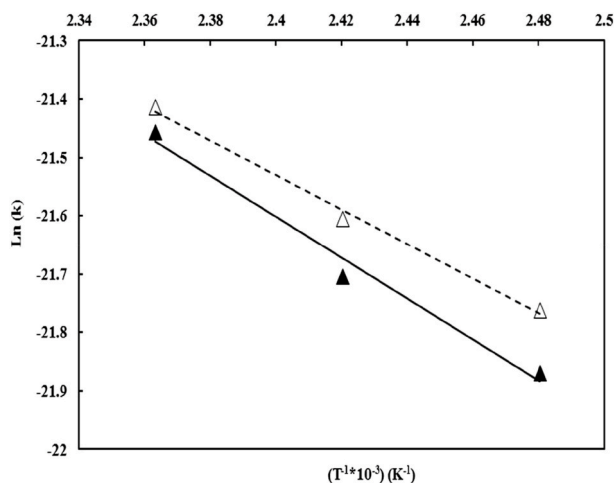


Fig. 4. Arrhenius plots obtained from the hydrolysis of PET with 2 M solutions of (a) PSSA (Δ), and (b) H_2SO_4 ($\blacktriangle$).

mechanism at the PET surface. Interestingly, while Fig. 7 shows that the activity of PSSA and H_2SO_4 in the hydrolysis of ethyl acetate are closer at lower temperatures (130 °C), this behavior was reversed in the case of the hydrolysis of PET, with the activities becoming closer at higher temperatures (150 °C). It is hypothesized that as the temperature increases, the solubility of PSSA in water decreases, thus increasing the interaction of PSSA with the surface of PET and improving the hydrolysis rate.

3.4. Mechanism for the acid hydrolysis of PET

In the acid hydrolysis of PET, protons (Brønsted acid sites) attack the polymer chains on the surface of the PET particles, resulting in a layer-by-layer reaction with no reaction occurring in the aqueous media. Since this reaction occurs at the polymer surface, any hydrophobic interactions between the catalyst and the PET particles could affect the reaction rate. The importance of hydrophobic interactions have been reported in the literature for acid-catalyzed reactions, such as the hydrolysis of cellulose [43] or acetalization reactions [61,62], and for the enzymatic hydrolysis of esters [63,64]. Specifically, these studies found that increasing the interaction of the catalyst with the surface resulted in an increased reaction rate.

To our knowledge, this is the first study that focuses on the hydrophobicity effect on the hydrolysis of PET using a recoverable acid catalyst. As noted above, PSSA exhibited a higher activity for the hydrolysis of PET than H_2SO_4 , particularly at higher concentrations. This difference can be attributed to a combination of improved surface wetting of PET by PSSA and the adsorption of PSSA onto the PET surface. Contact angle measurements with PSSA and H_2SO_4 aqueous solutions on the PET surface were used to evaluate the interaction between PSSA and the hydrophobic surface [65]. Fig. 8 shows images of droplets of 0.5 M, 1 M, and 2 M aqueous solutions of PSSA and H_2SO_4 on the surface of PET films.

As seen in Fig. 8, the droplets of PSSA solutions exhibited greater wetting of the PET surface than the H_2SO_4 solutions, particularly at higher concentrations. This is consistent with literature reports on PSSS wetting an alumina surface due to the hydrophobic nature of the polystyrene backbone and improvements in the configurational entropy of PSSS that drives the surface adsorption [66]. This adsorption is dependent upon the nature of the surface, the concentration of PSSA, and the

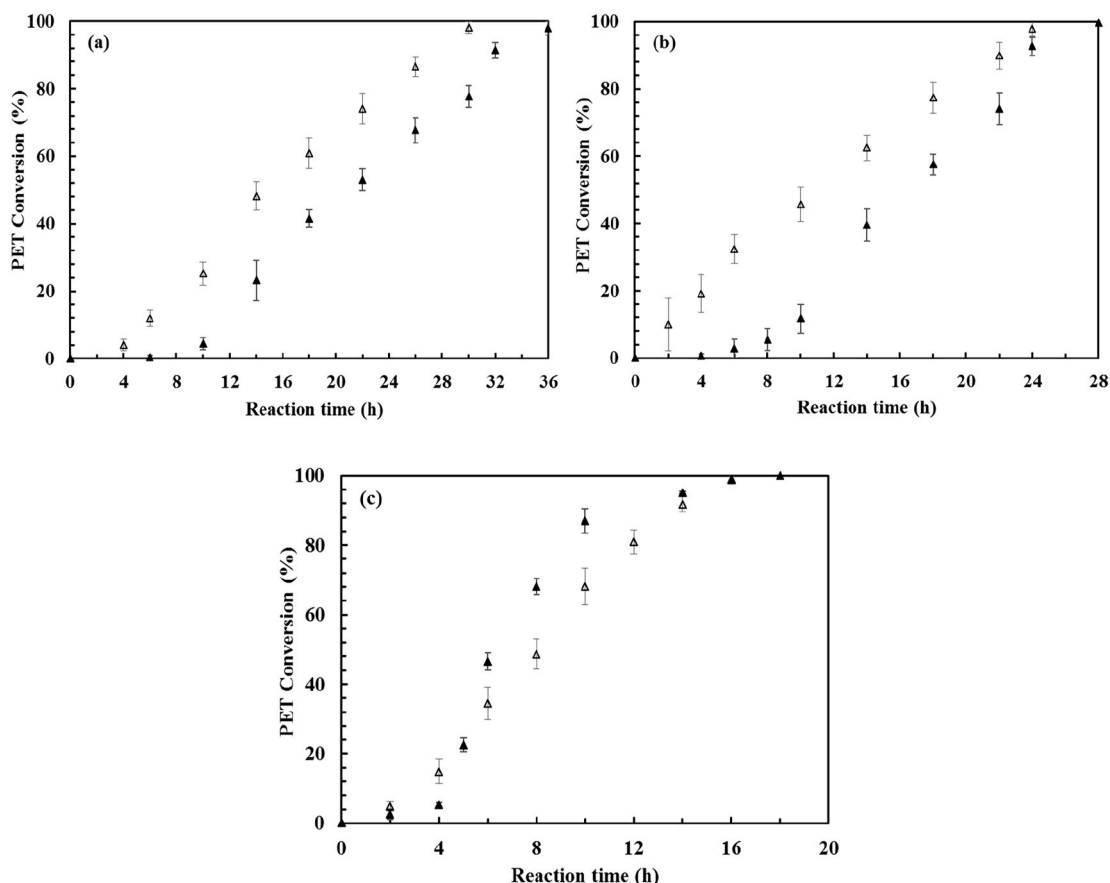


Fig. 5. PET conversions (%) obtained with (a) 1 M, (b) 2 M, and (c) 4 M solutions of PSSA (Δ) and H_2SO_4 ($\blacktriangle$) at $T_{\text{reaction}} = 150$ °C.

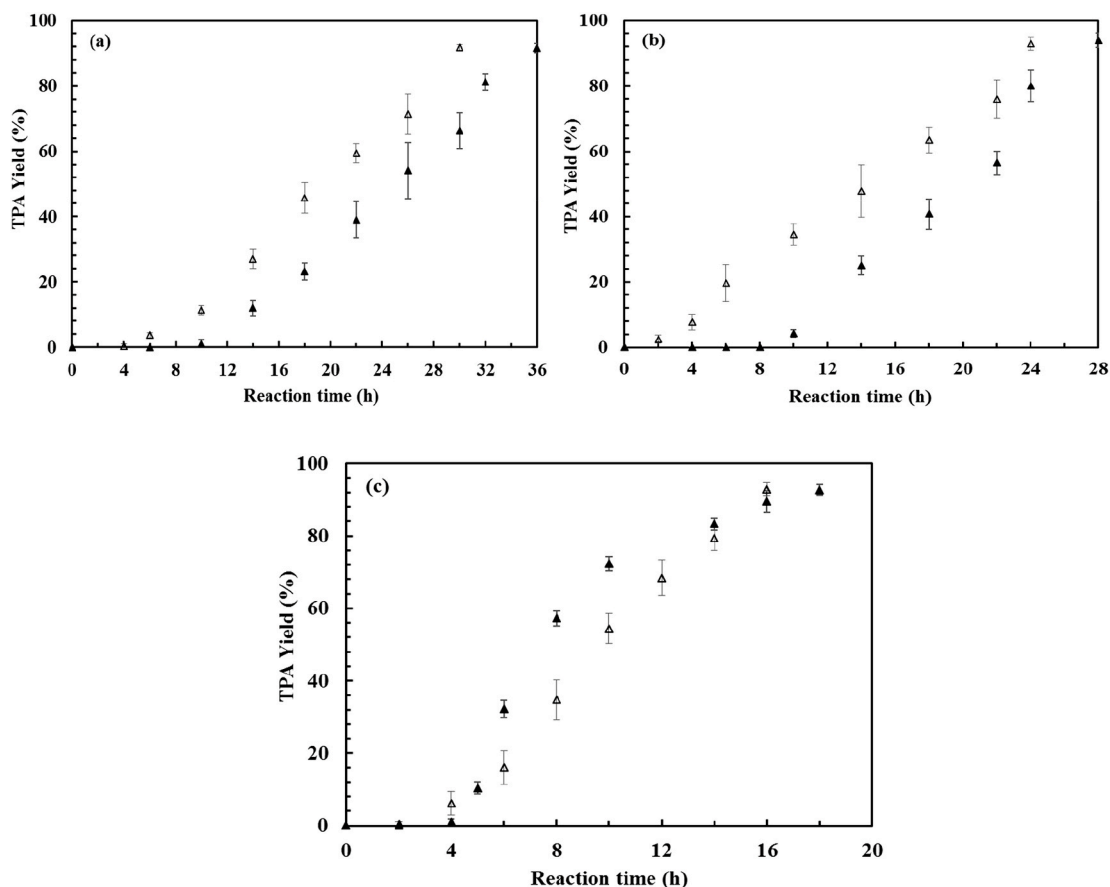


Fig. 6. TPA yields (%) obtained with (a) 1 M, (b) 2 M, and (c) 4 M solutions of PSSA (Δ) and H₂SO₄ ($\blacktriangle$) at $T_{\text{reaction}} = 150\text{ }^{\circ}\text{C}$.

Table 2

Apparent reaction rate constants (K) vs catalyst concentration (H^+), based on the shrinking-core model and TPA recovery with PSSA and H₂SO₄ catalysts at $T_{\text{reaction}} = 150\text{ }^{\circ}\text{C}$.

$T_{\text{reaction}} = 150\text{ }^{\circ}\text{C}$	PSSA	H ₂ SO ₄
Catalyst Concentration (M)	$K\text{ (h}^{-1}\text{)} \cdot (10^2)$	$K\text{ (h}^{-1}\text{)} \cdot (10^2)$
1	1.52	1.38
2	1.73	1.66
4	3.66	3.72

configuration of PSSA in solution. Additionally, this will lead into a higher effective local concentration of Brønsted acid sites in the vicinity of the PET surface relative to H₂SO₄ at a given solution concentration. The increased wetting by PSSA solutions is reflected in a decrease in the contact angle with increasing concentrations, as shown in Fig. 9. While the contact angle for H₂SO₄ solutions with different concentrations ranged between 78 and 86°, the contact angles with PSSA solutions were considerably lower, particularly at higher molarities.

3.5. Reutilization experiments with PSSA catalyst

PSSA was recovered by evaporation from the filtrate after 14 h of reaction at 150 °C with a 2 M solution of PSSA to examine its reusability for acid hydrolysis of PET. Interestingly, the structure of the PSSA catalyst showed to be constant after five catalytic cycles (Fig. S4 to Fig. S6 in the Supplementary Material). Fig. 10 presents the reutilization results obtained after five cycles with a 2 M solution of PSSA catalyst at 150 °C for 14 h, including PET conversions and the TPA yields. As can be seen in Fig. 10, the PSSA maintained its activity during five cycles without significant loss of activity. Fig. S7 shows the PSSA recovery yield (% of mass recovered) after every reaction round, which was close to 100% in all cases. Additionally, the acidity of the recovered PSSA was evaluated after every cycle by titration with a 0.01 N NaOH solution. The titration results (Fig. 11) confirmed that PSSA maintained the number of Brønsted acid sites after every reaction cycle.

4. Conclusions

Poly (4-styrenesulfonic acid) (PSSA) was investigated as a reusable and recoverable catalyst for the acid hydrolysis of PET. The catalytic activity of PSSA was compared with that of a conventional mineral acid

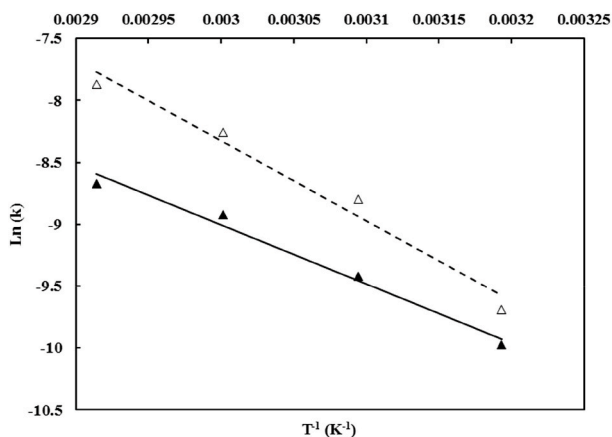


Fig. 7. Arrhenius plots obtained from the hydrolysis of ethyl acetate with (a) PSSA (Δ), and (b) H₂SO₄ ($\blacktriangle$).

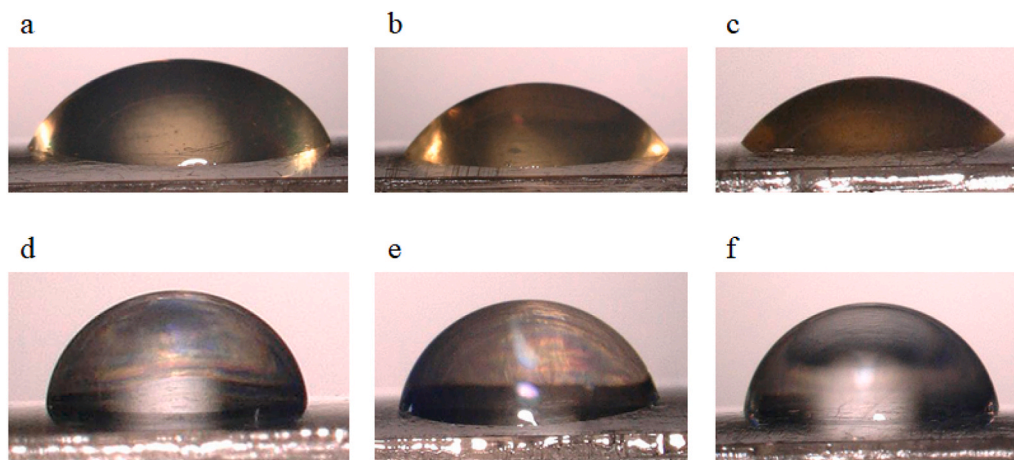


Fig. 8. (a–c) 0.5 M, 1 M, and 2 M solutions of PSSA on the PET surface, respectively, and (d–f) 0.5 M, 1 M, and 2 M solutions of H_2SO_4 on the PET surface, respectively.

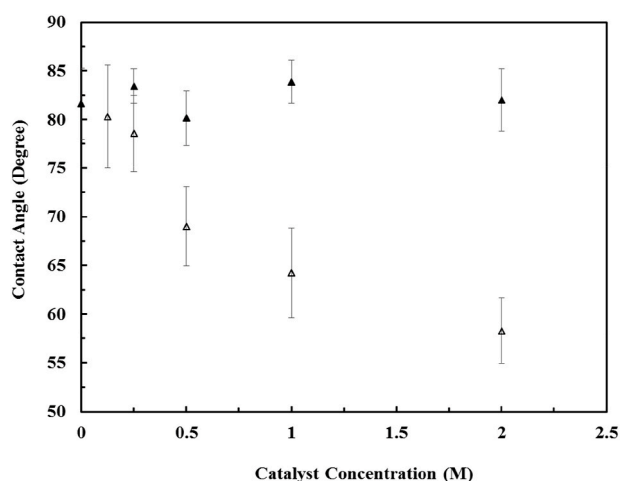


Fig. 9. Contact angle measurements obtained with different concentrations of PSSA (Δ) and H_2SO_4 ($\blacktriangle$) aqueous solutions on the PET surface.

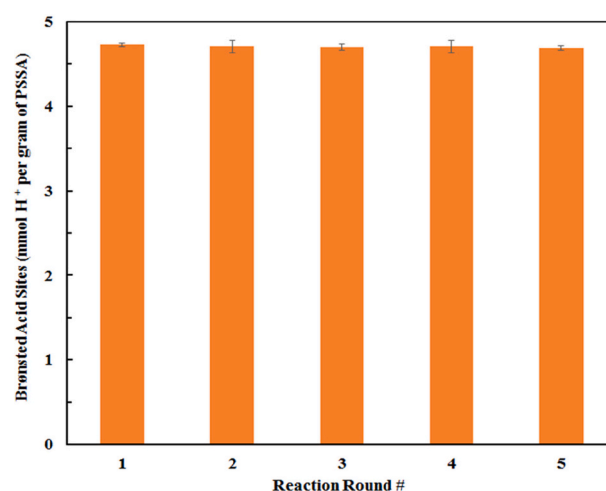


Fig. 11. Titration results obtained from the recovered PSSA after every reaction cycle.

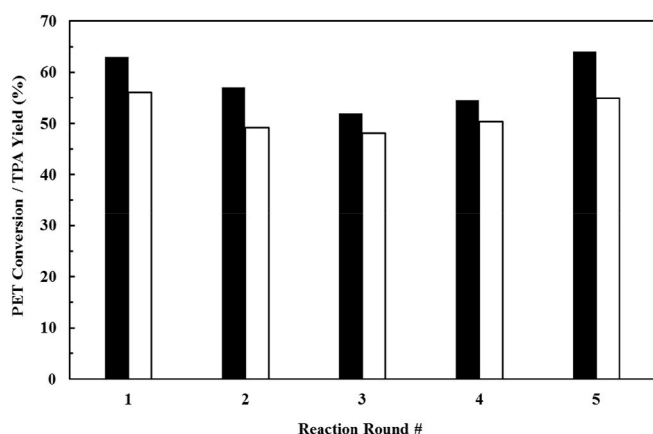


Fig. 10. Reutilization results obtained with a 2 M solution of PSSA: PET conversion ($\blacksquare$) and TPA yield ($\square$). Every reaction was run at $150\text{ }^\circ\text{C}$ for 14 h.

catalyst (H_2SO_4) for the depolymerization of PET. The PSSA systems exhibited much shorter induction times and slightly higher activity than H_2SO_4 at similar reaction conditions.

The shrinking-core model provided a good fit to the kinetics data and

allowed the determination of the specific reaction rate constants and the activation energies. While hydrolysis studies of a small model molecule (ethyl acetate) showed a lower activation energy with H_2SO_4 (39.6 vs 53.8 kJ mol^{-1} with PSSA), PSSA was more active (24.8 kJ mol^{-1}) than H_2SO_4 (29.1 kJ mol^{-1}) in the hydrolysis of PET.

The increased reaction rate and the reduction of the induction time with PSSA was attributed to improved surface wetting combined with adsorption of PSSA on PET surface because of the hydrophobic region of PSSA. This improved wetting was confirmed by contact angle measurements of PSSA and H_2SO_4 aqueous solutions on PET surfaces. While the contact angle of the H_2SO_4 solutions remained constant and near the value for water, there was a decrease in the contact angle for PSSA solutions with increasing concentrations. These lower contact angles with the PSSA solutions are consistent with reports in the literature and are attributed to the hydrophobic nature of the polystyrene backbone with improved interaction with PET particles. This should lead into a higher local concentration of acid sites near the PET surface, and consequently, higher activity in the hydrolysis reaction. Further, the reutilization and titration results confirmed that PSSA is a reusable and recoverable catalyst, which is another advantage from an environmental point of view with respect to H_2SO_4 .

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.polymer.2021.123620>.

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